

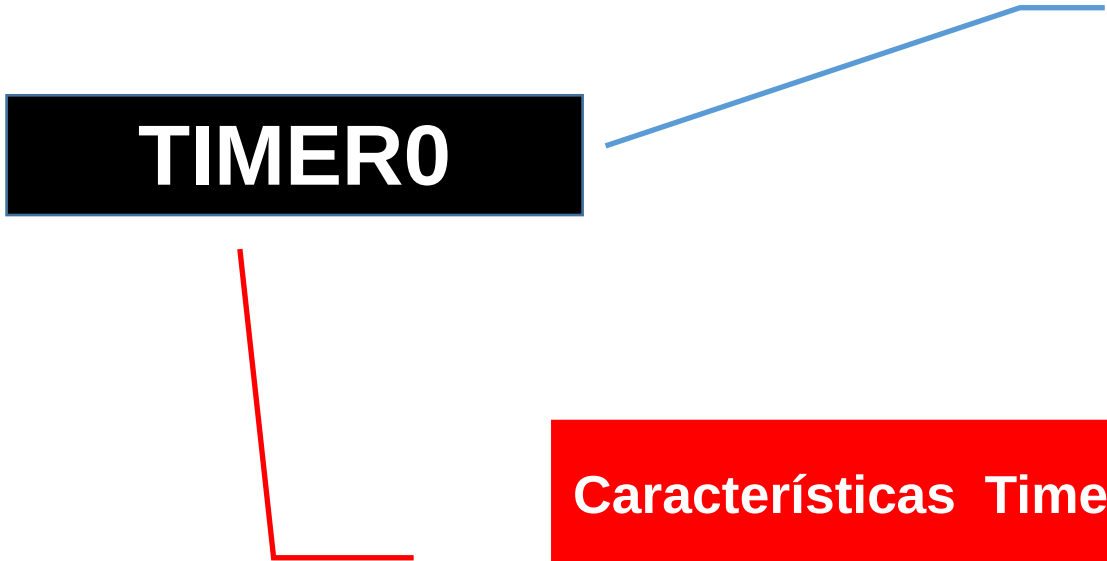


TIMERs

TEMPORIZADORES DE USOS GENERALES

PIC16F887: Timer 0 – Timer 1 – Timer 2

TIMER0

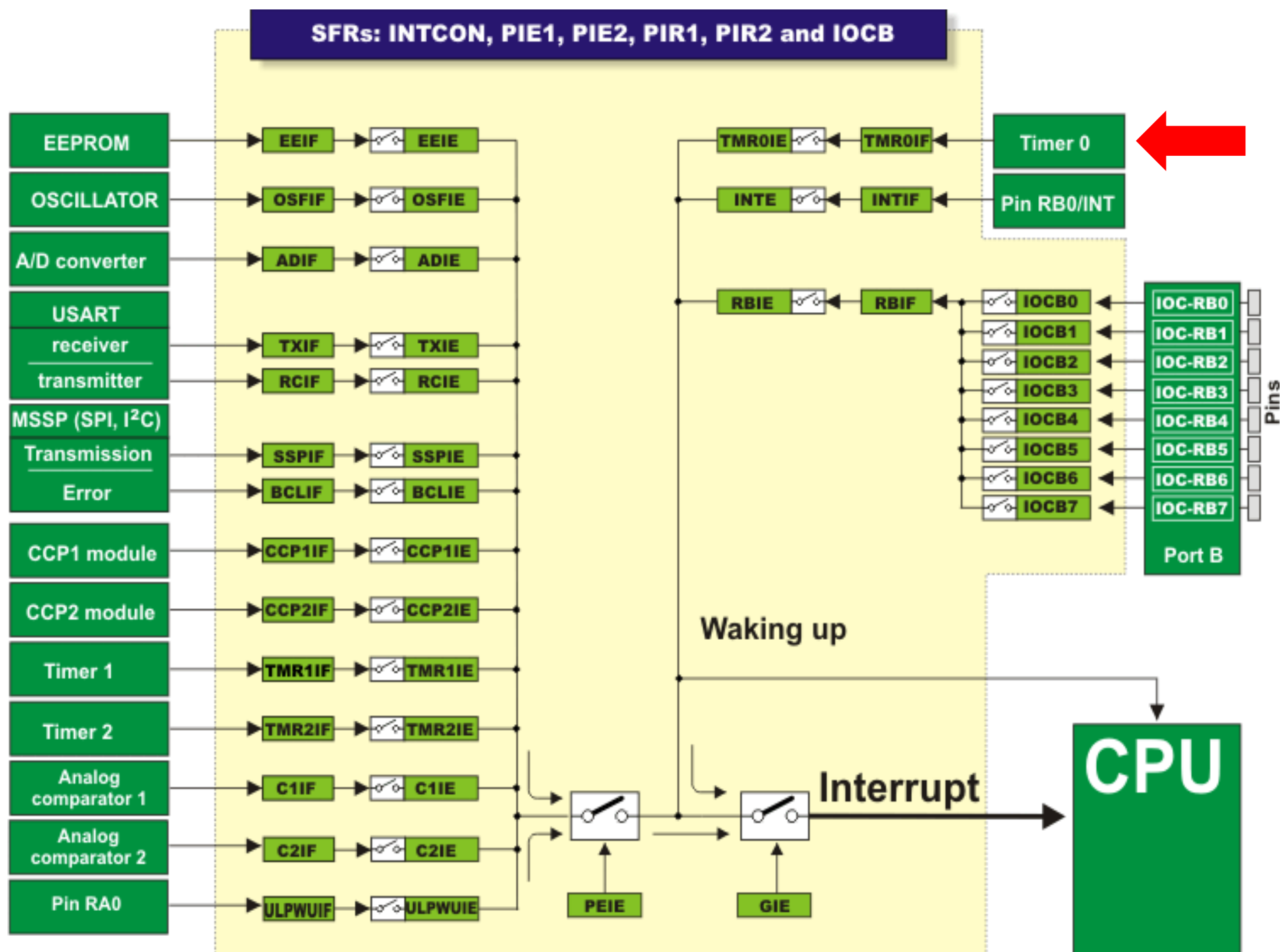


Usos:

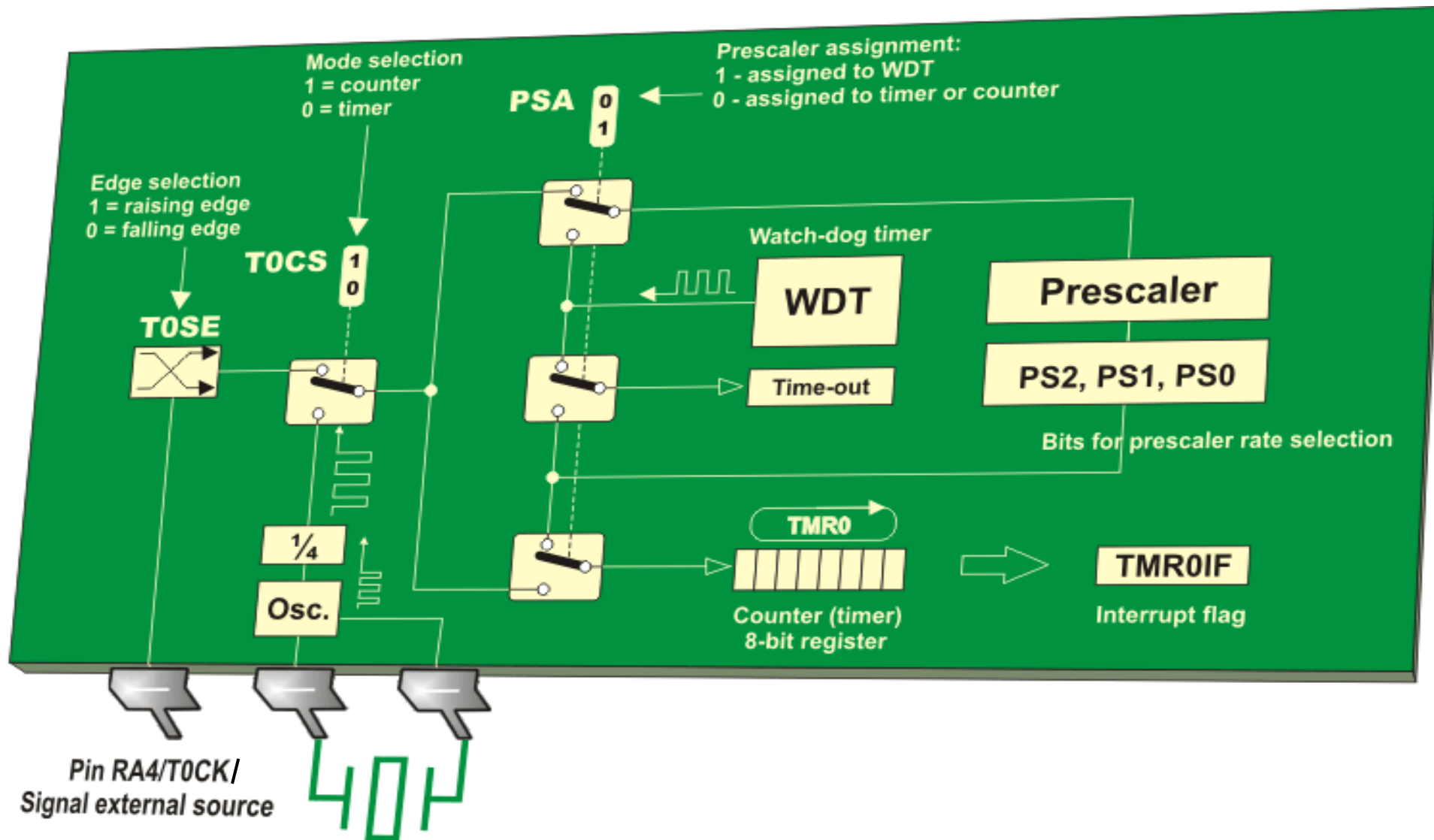
- Generar pulsos de duración programable
- Medición de tiempo
- Contar pulsos externos

Características Timer 0:

- ✓ Temporizador/Contador de 8 bits
- ✓ Prescaler de 3 bits (8 valores) compartido con el WDT
- ✓ Fuente de Reloj interna o externa (elegible por programa)
- ✓ Interrupción disponible (en desbordamiento)
- ✓ Flanco de Reloj externo programable

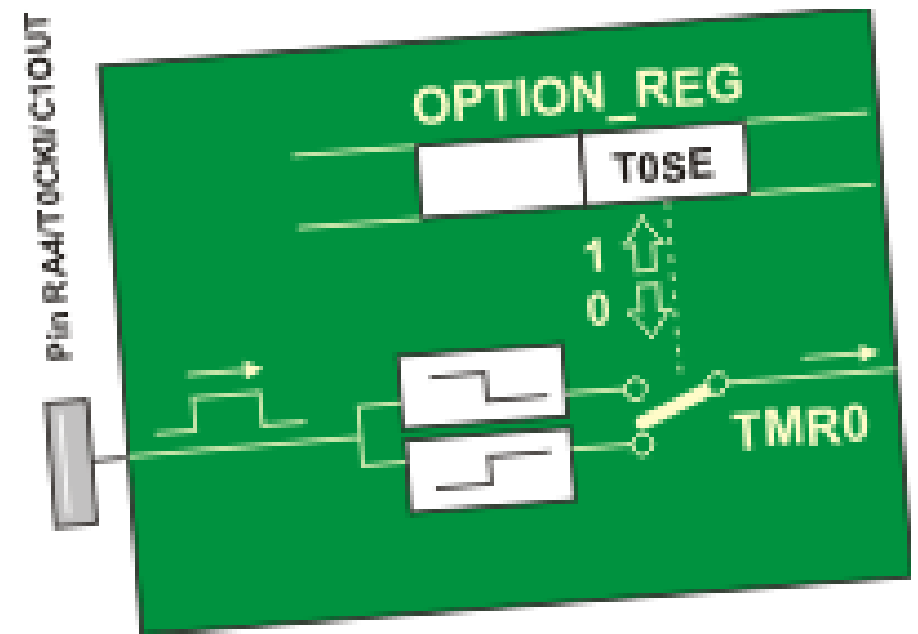
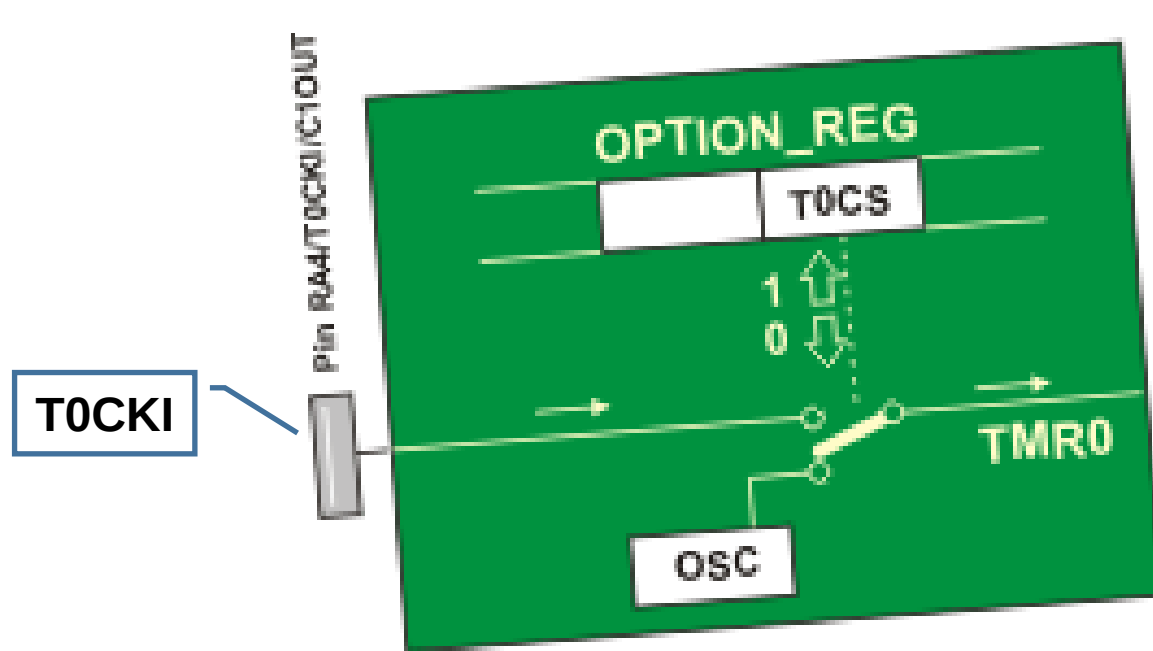


TIMER0:



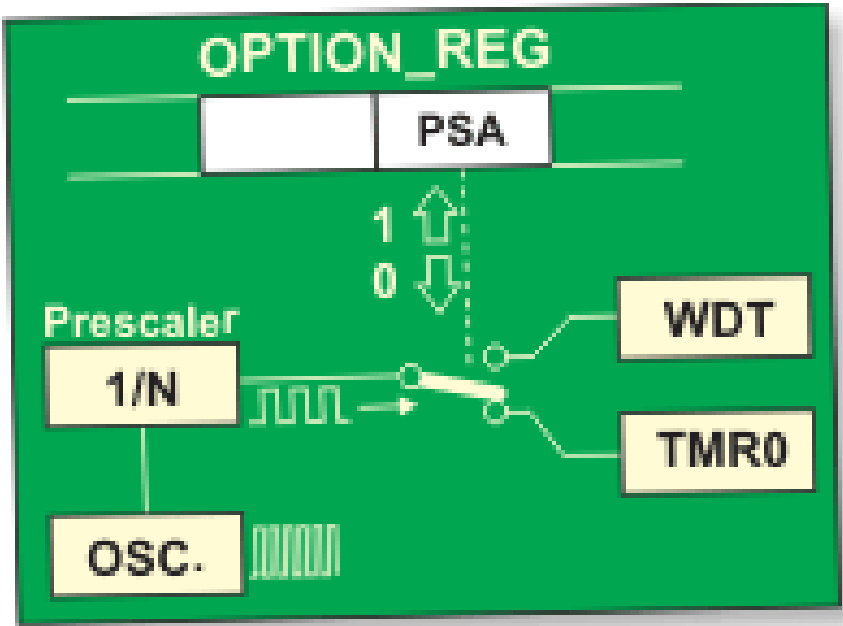
Registro OPTION_REG

OPTION	R/W (1)	R/W (1)	R/W (1)	R/W (1)	R/W (1)	R/W (1)	R/W (1)	Features
	RBPU	INTEDG	T0CS	T0SE	PSA	PS2	PS1	Bit name
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0



Registro OPTION_REG

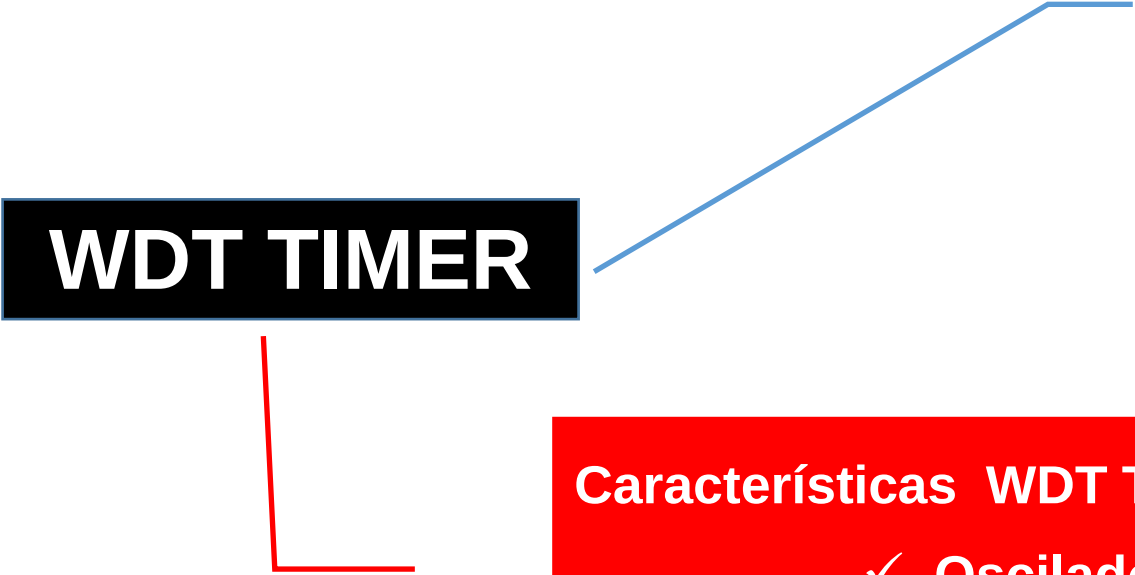
OPTION	R/W (1)	R/W (1)	R/W (1)	R/W (1)	R/W (1)	R/W (1)	R/W (1)	R/W (1)	Features
	RBPU	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	Bit name
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	



PS2	PS1	PS0	TMR0	WDT
0	0	0	1: 2	1: 1
0	0	1	1: 4	1: 2
0	1	0	1: 8	1: 4
0	1	1	1:16	1: 8
1	0	1	1:64	1:32
1	1	0	1: 128	1:64
1	1	1	1: 256	1: 128

TEMPORIZADOR DEDICADO: WATCHDOG

WDT TIMER



Usos:

- Provoca Reset ante fallas del programa
- Saca (wake up) al μ P del modo dormido (SLEEP)

Características WDT Timer:

- ✓ Oscilador interno RC
- ✓ Prescaler de 3 bits compartido con TM0
- ✓ Su uso o no se define en configuración: `_WDTE_OFF`
- ✓ Reset por desbordamiento $\overline{TO}$ en STATUS
- ✓ Para evitar desbordamiento `CLRFWDT`
- ✓ Tiempo: 10/16/29 ms (típico sin prescaler = 16 ms)
Tiempo máximo con prescaler = ?

Pin RA4/T0CK

PSA 0

T0CS 1 0

T0SE

1/4

Osc.

WDT

Time-out

Prescaler

PS2, PS1, PS0

TMR0

TMR0IF

Pin RA4/T0CK

PSA 1

T0CS 1 0

T0SE

1/4

Osc.

WDT

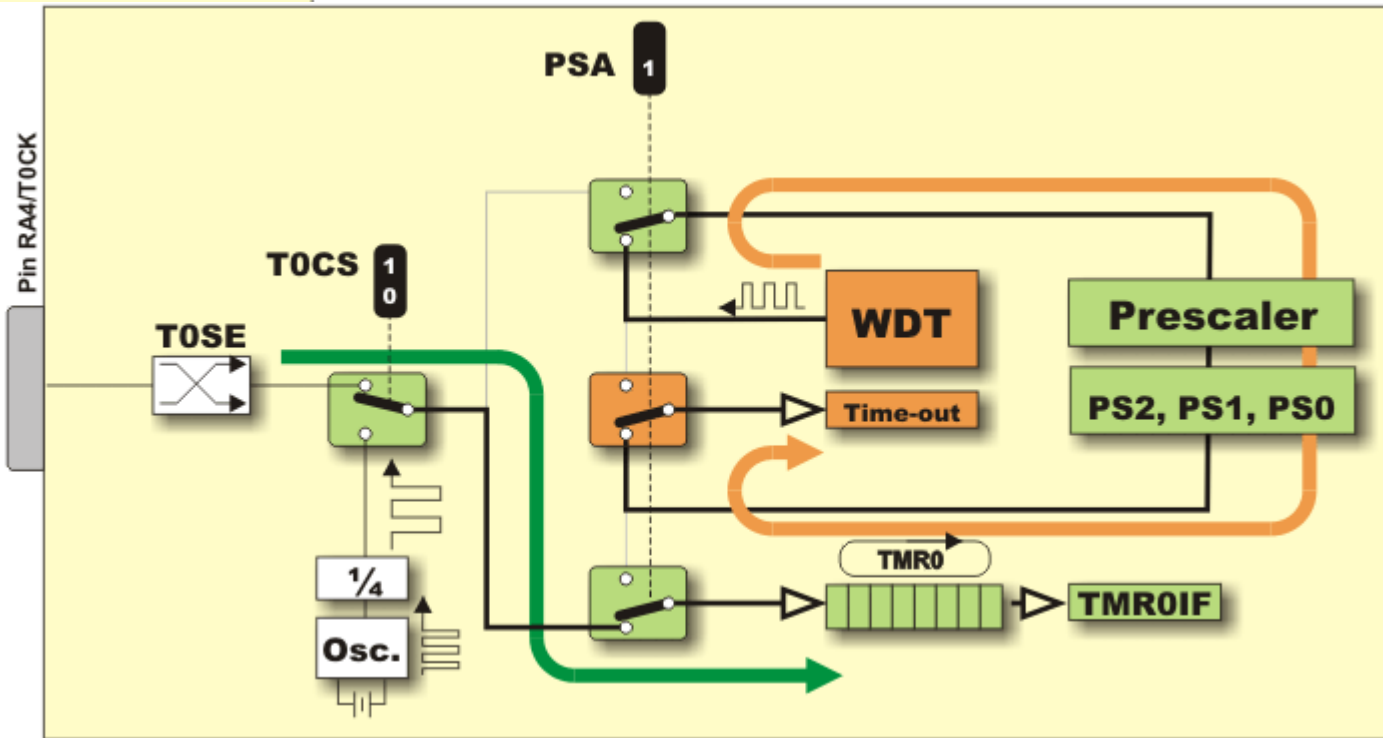
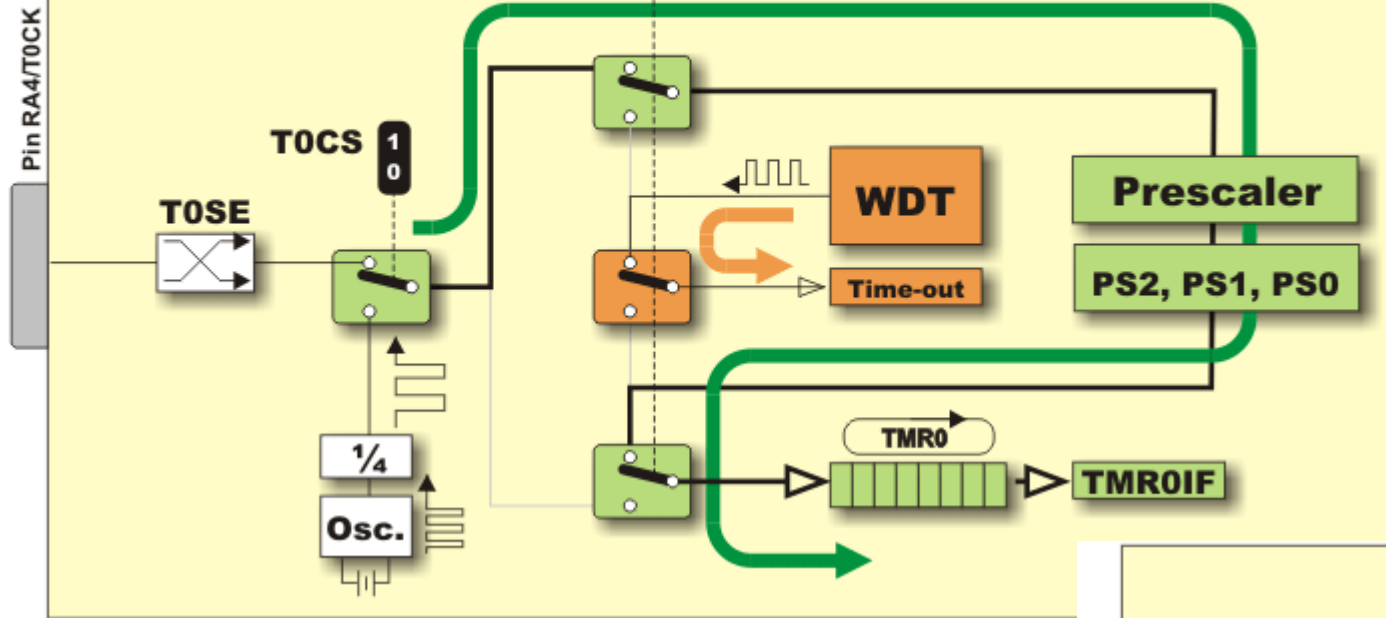
Time-out

Prescaler

PS2, PS1, PS0

TMR0

TMR0IF



CONSIDERACIONES AL USAR TMR0 MODULE

- ✓ Escribir en el registro TMR0 no inicia el conteo de inmediato sino con dos ciclos de instrucción de retraso
- ✓ Si el uP está en modo dormido el oscilador se apaga, no hay pulsos y no se produce interrupción. No puede sacar al procesador de un SLEEP.
- ✓ Si se usa como contador sin prescaler la duración tanto en bajo como en alto del pulso debe ser $0,5 t_i + 20 \text{ nS}$.
- ✓ Cuando se usa como un contador de reloj externo con prescaler la duración de pulso mínima o una pausa entre dos pulsos es de 10 nS
- ✓ Al cambiar la asignación del prescaler de TMR0 al WDT se debe ejecutar una secuencia de instrucciones para evitar que se resetee. Ver la secuencia en el Datasheet "TIMER0 MODULE"
- ✓ Al cambiar la asignación del prescaler de WDT a TMR0 se debe ejecutar una secuencia de instrucciones para evitar que se resetee. Ver la secuencia en el Datasheet "TIMER0 MODULE"

Registro INTCON

INTCON	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (x)	Features
	GIE	PEIE	TOIE	INTE	RBIE	TOIF	INTF	RBIF	Bit name
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	

Legend

R/W	Readable/Writable bit
(0)	After reset, bit is cleared
(x)	After reset, bit is unknown

$$\text{Tiempo} = ((256 - \text{TMR0}) * \text{PS} + 2) * t_i$$

Calcular el tiempo máximo con $f_{\text{osc}} = 4 \text{ MHz}$

$$t_i = 4 \times t_c = 4 \times 0,25 \mu\text{s} = 1 \mu\text{s}$$

$$\text{PS} = 256 \Rightarrow \text{PS2}, \text{PS1}, \text{PS0} = 1, 1, 1$$

$$\text{Tmax} = ((256 - 0) \times 256 + 2) \times 1 \mu\text{s} = 65.538 \mu\text{s} = 65,53 \text{ ms}$$

$$\text{Tmax} \approx 256 \times 256 \times t_i$$

Si se desea un tiempo de 500 ms $\Rightarrow$ $T_{\text{total}} = 10 * 50 \text{ ms}$ colocando el TMR0 en 50 ms

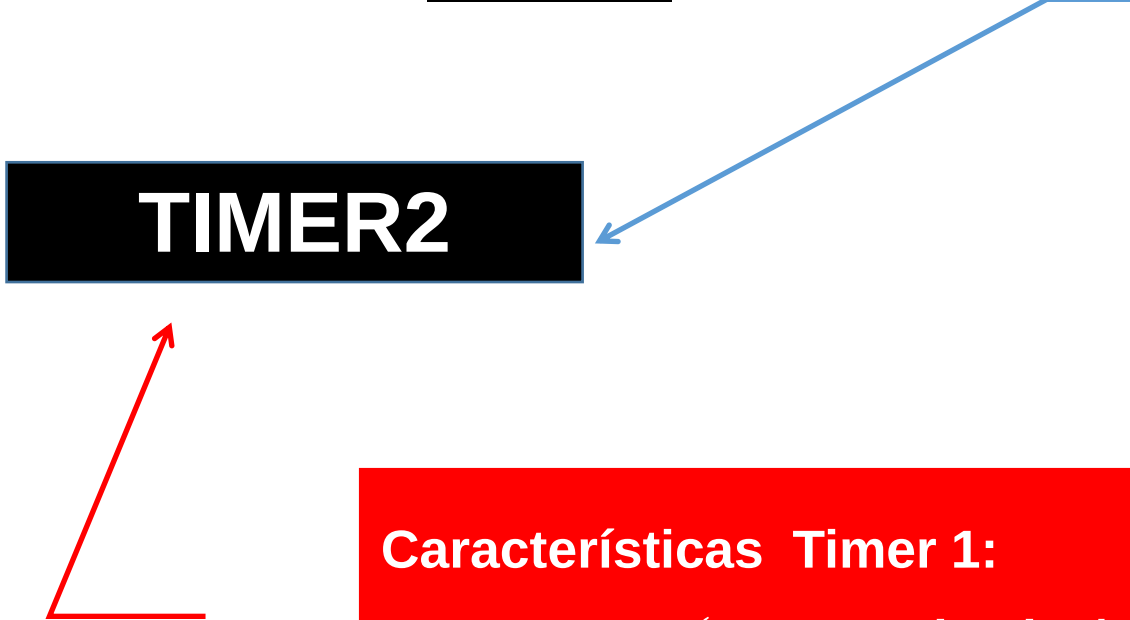
TABLE 5-1: SUMMARY OF REGISTERS ASSOCIATED WITH TIMER0

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
TMRO	Timer0 Module Register									
INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF	xxxx xxxx	uuuu uuuu
OPTION_REG	$\overline{\text{RBPU}}$	INTEDG	T0CS	T0SE	PSA	PS2	PS1	PS0	0000 000x	0000 000x
TRISA	TRISA7	TRISA6	TRISA5	TRISA4	TRISA3	TRISA2	TRISA1	TRISA0	1111 1111	1111 1111
									1111 1111	1111 1111

Legend: – = Unimplemented locations, read as '0', u = unchanged, x = unknown. Shaded cells are not used by the Timer0 module.

TEMPORIZADORES DE USOS GENERALES

PIC16F887: Timer 0 – Timer 1 – Timer 2



TIMER2

Usos:

- Generar pulsos de duración programable
- Medición de tiempo
- Contar pulsos externos

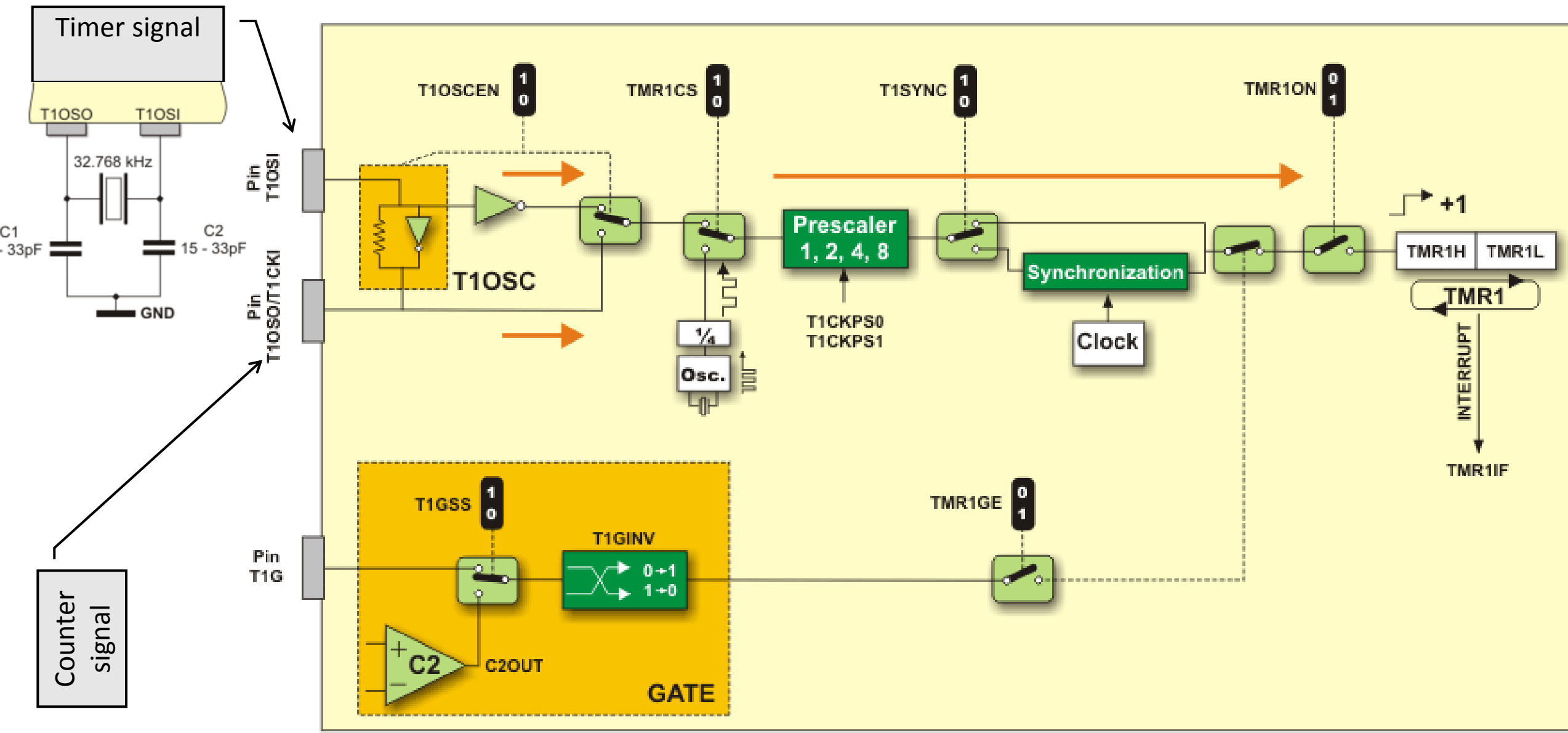
Características Timer 1:

- ✓ Temporizador/Contador de 16 bits (TMR1H:TMR1L)
- ✓ Prescaler con valores: 1,2,4 y 8 independiente
- ✓ Oscilador opcional de baja potencia (LP)
- ✓ Fuente de Reloj interna o externa (elegible por programa)
- ✓ Interrupción disponible (en desbordamiento)
- ✓ Puerta para cronometrar eventos interna o externa
- ✓ Puede interrumpir después de un SLEEP

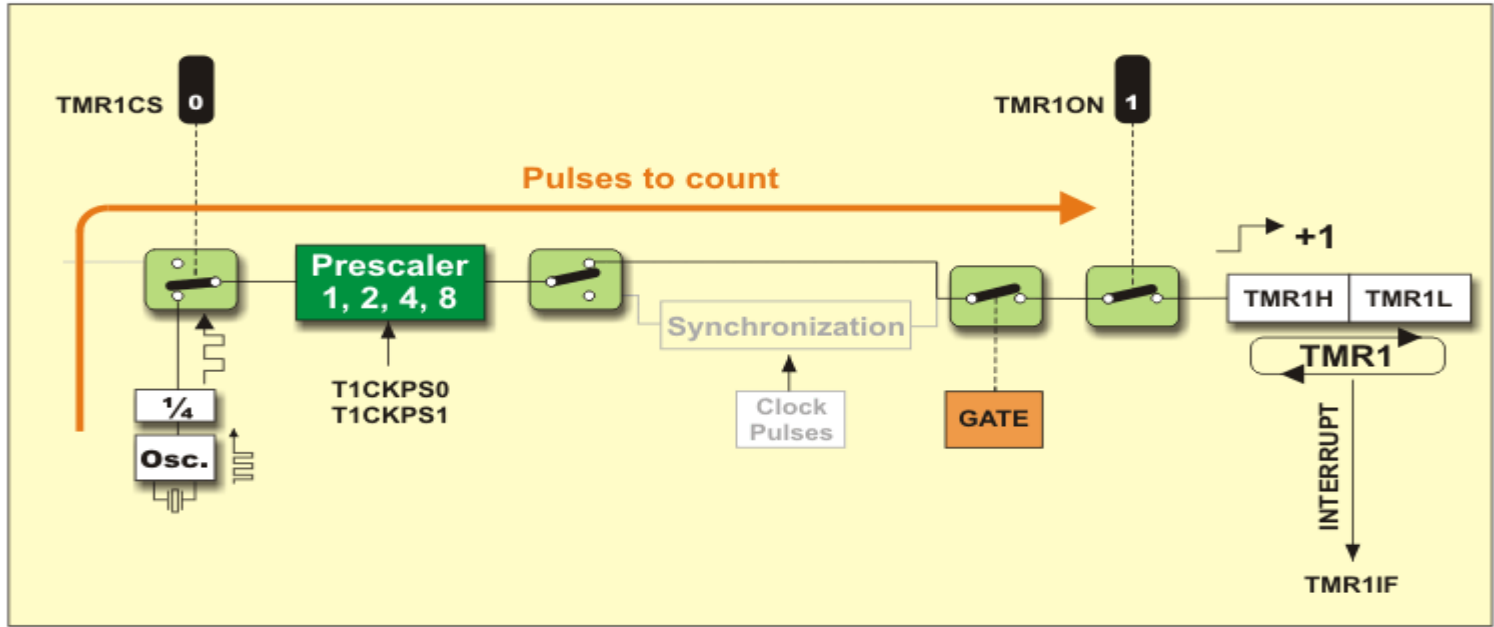
SFRs: INTCON, PIE1, PIE2, PIR1, PIR2 and IOCB



TIMER1:

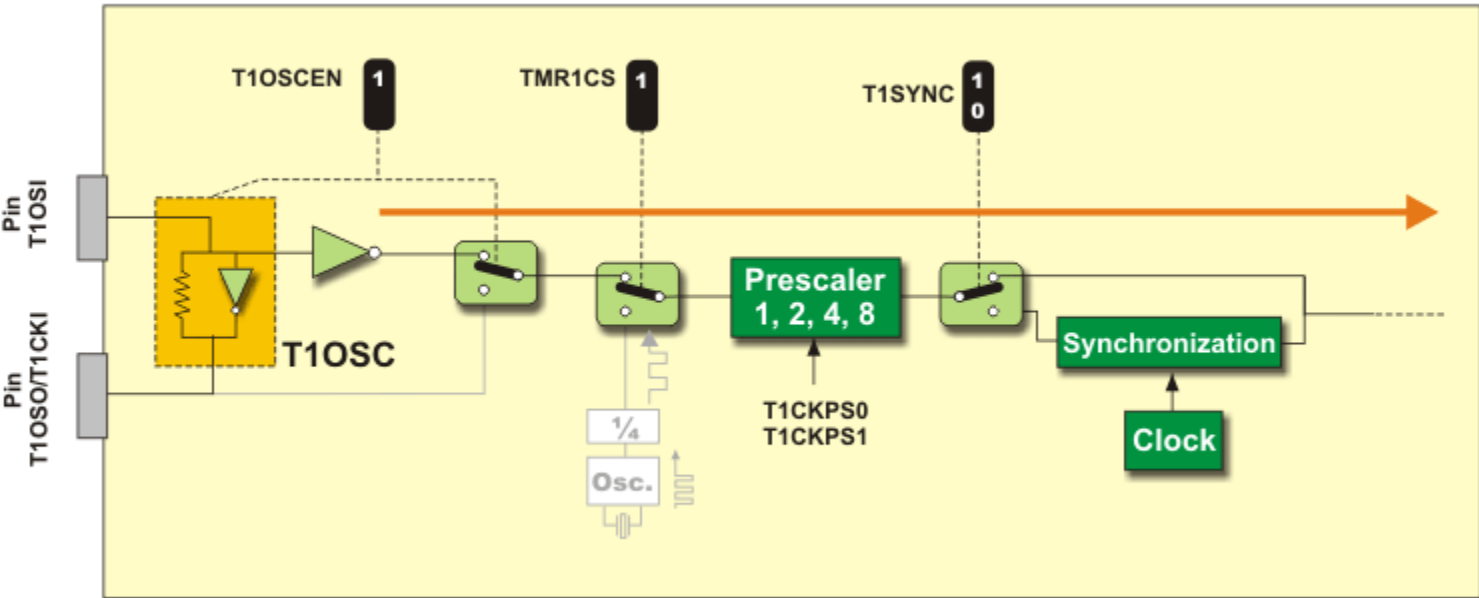


Fuente del reloj del TMR1:



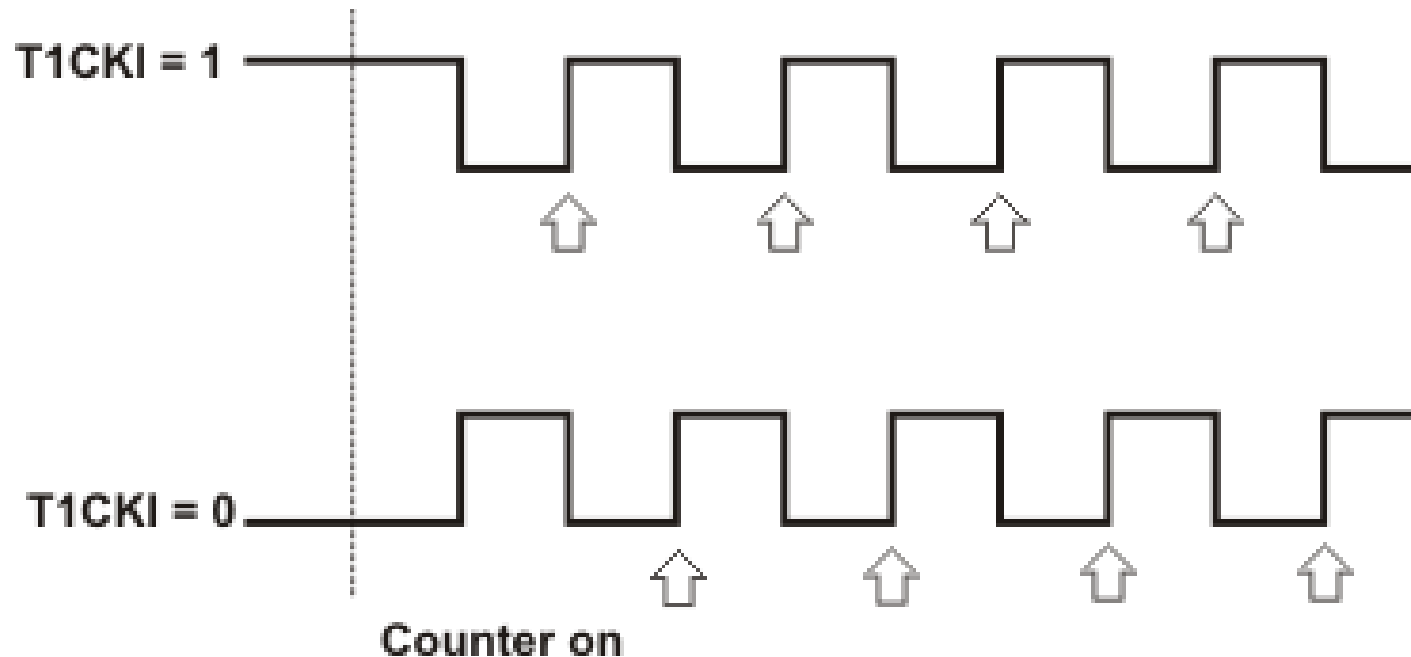
INTERNA
(solo timer)

EXTERNA
(solo timer)



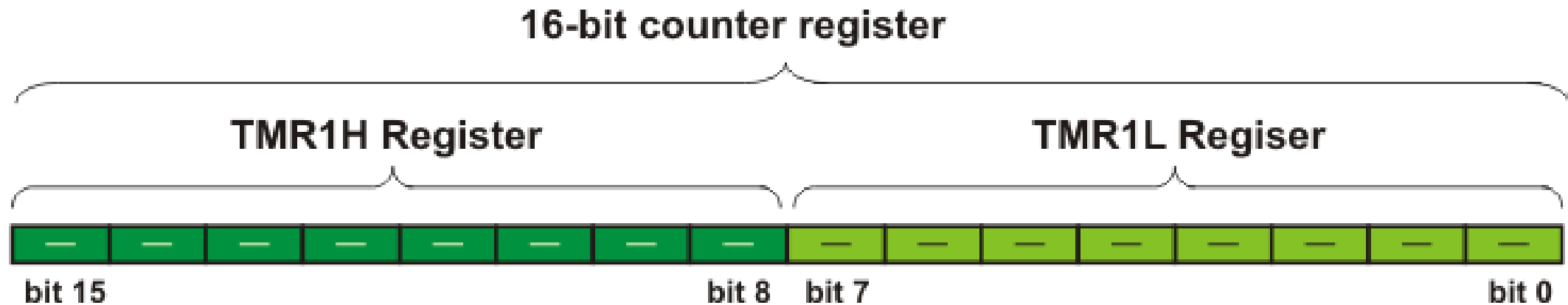
El modo contador en TMR1:

- ✓ cuenta solo pulsos por flanco ascendente
- ✓ se activa la cuenta por flanco descendente
- ✓ para que cuente antes debió existir un flanco descendente



Registros para programación del Timer 1:

Cuenta hasta 65.535 pulsos en una sola corrida



Nota: leer o escribir en estos registros limpia el prescaler

	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	Features	
T1CON	T1GINV	TMR1GE	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNC	TMR1CS	TMR1ON	Bit name
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	

Legend

R/W (0) Readable/Writable bits
After reset, bit is cleared

- bit 7 **T1GINV**: Timer1 Gate Invert bit⁽¹⁾
1 = Timer1 gate is active-high (Timer1 counts when gate is high)
0 = Timer1 gate is active-low (Timer1 counts when gate is low)
- bit 6 **TMR1GE**: Timer1 Gate Enable bit⁽²⁾
If TMR1ON = 0:
This bit is ignored
If TMR1ON = 1:
1 = Timer1 is on if Timer1 gate is not active
0 = Timer1 is on
- bit 5-4 **T1CKPS<1:0>**: Timer1 Input Clock Prescale Select bits

T1CKPS1/0	PRESCALER
0 0	1: 1
0 1	1: 2
1 0	1: 4
1 1	1: 8

T1CON	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	Features
	T1GINV	TMR1GE	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNC	TMR1CS	TMR1ON
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
								Bit name

- bit 3 **T1OSCEN:** LP Oscillator Enable Control bit
1 = LP oscillator is enabled for Timer1 clock
0 = LP oscillator is off
- bit 2 **T1SYNC:** Timer1 External Clock Input Synchronization Control bit
TMR1CS = 1:
1 = Do not synchronize external clock input
0 = Synchronize external clock input
TMR1CS = 0:
This bit is ignored. Timer1 uses the internal clock
- bit 1 **TMR1CS:** Timer1 Clock Source Select bit
1 = External clock from T1CKI pin (on the rising edge)
0 = Internal clock (FOSC/4)
- bit 0 **TMR1ON:** Timer1 On bit
1 = Enables Timer1
0 = Stops Timer1

- Note 1:** T1GINV bit inverts the Timer1 gate logic, regardless of source.
- 2:** TMR1GE bit must be set to use either $\overline{T1G}$ pin or C2OUT, as selected by the T1GSS bit of the CM2CON1 register, as a Timer1 gate source.

TABLE 6-1: SUMMARY OF REGISTERS ASSOCIATED WITH TIMER1

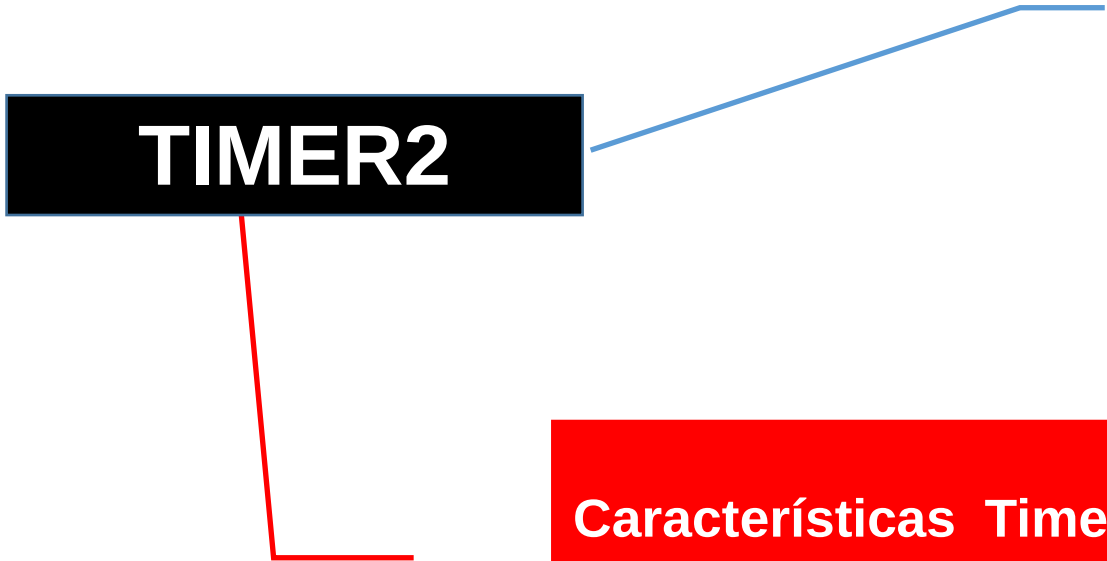
Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
CM2CON1	MC1OUT	MC2OUT	C1RSEL	C2RSEL	—	—	T1GSS	C2SYNC	0000 --10	0000 --10
INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF	0000 000x	0000 000x
PIE1	—	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE	-000 0000	-000 0000
PIR1	—	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF	-000 0000	-000 0000
TMR1H	Holding Register for the Most Significant Byte of the 16-bit TMR1 Register								xxxxx xxxxx	uuuu uuuu
TMR1L	Holding Register for the Least Significant Byte of the 16-bit TMR1 Register								xxxxx xxxxx	uuuu uuuu
T1CON	T1GINV	TMR1GE	T1CKPS1	T1CKPS0	T1OSCEN	T1SYNC	TMR1CS	TMR1ON	0000 0000	uuuu uuuu

Legend: x = unknown, u = unchanged, – = unimplemented, read as '0'. Shaded cells are not used by the Timer1 module.

TEMPORIZADORES DE USOS GENERALES

PIC16F887: Timer 0 – Timer 1 – Timer 2

TIMER2



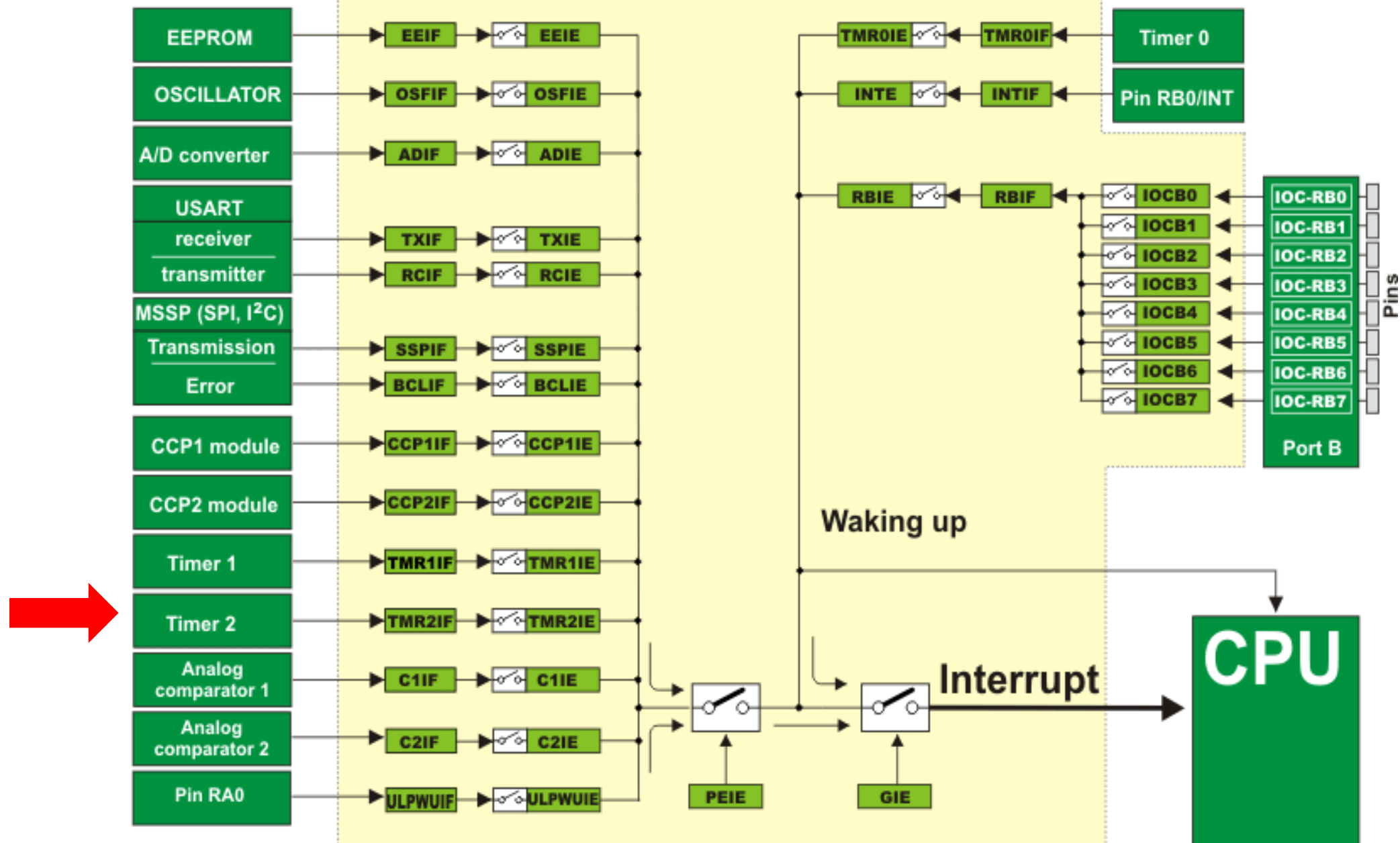
Usos:

- Generar pulsos de duración programada
- Temporiza en módulo USART
- Temporiza en módulo PWM

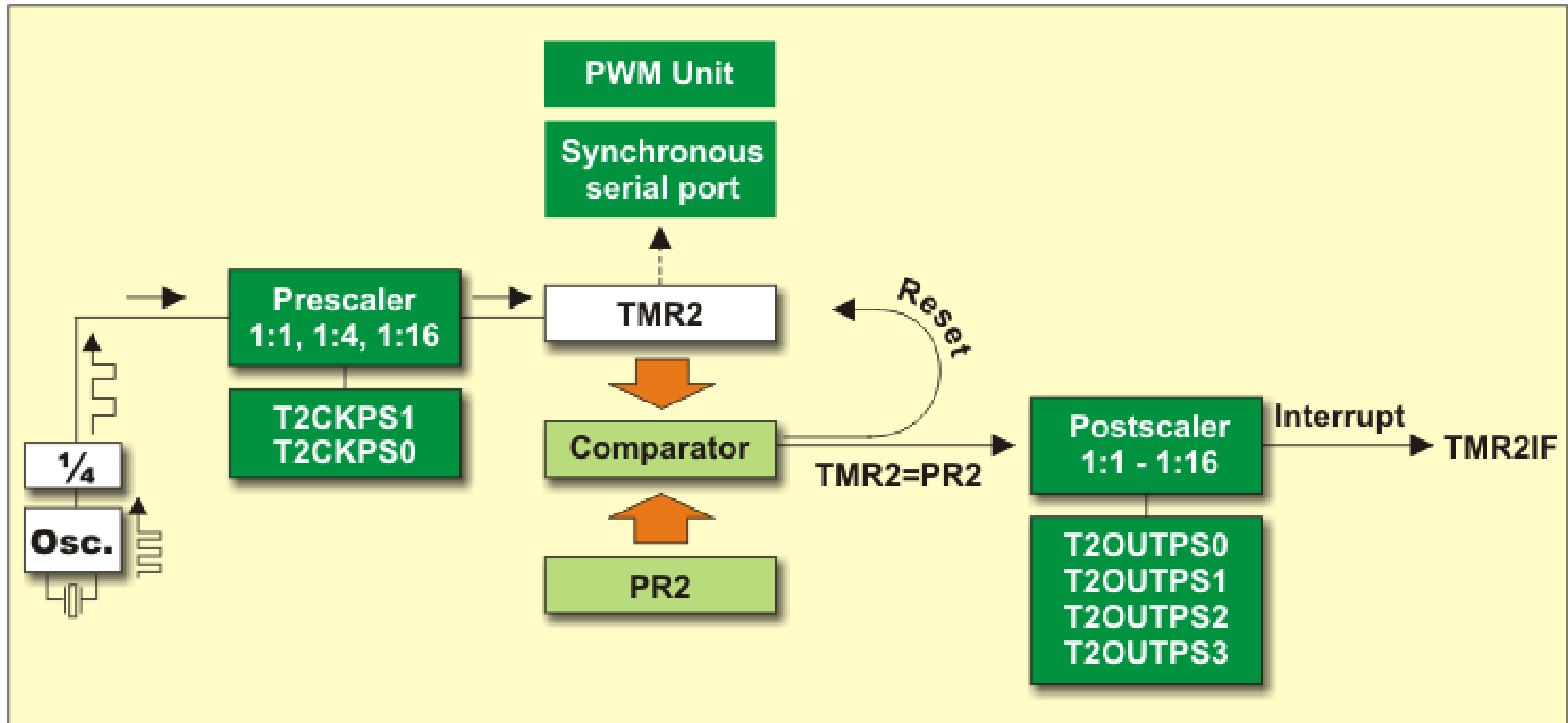
Características Timer 0:

- ✓ Temporizador de 8 bits (TMR2)
- ✓ Período de referencia de 8 bits (PR2)
- ✓ Prescaler de 2 bits (3 valores)
- ✓ Postescaler de 4 bits (16 bits)
- ✓ Fuente de Reloj interna e igual a ti ($F_{osc}/4$)
- ✓ Interrupción cuando $TMR2=PR2$

SFRs: INTCON, PIE1, PIE2, PIR1, PIR2 and IOCB



TIMER2:



Registro para programación del Timer 2:

								Features
								Bit name
T2CON	-	TOUTPS3	TOUTPS2	TOUTPS1	TOUTPS0	TMR2ON	T2CKPS1	T2CKPS0
	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
R/W (0)		R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	R/W (0)	

Legend

- Bit is unimplemented
- R/W Readable/Writable bit
- (0) After reset, bit is cleared

bit 6-3	TOUTPS<3:0> : Timer2 Output Postscaler Select bits
	0000 = 1:1 Postscaler
	0001 = 1:2 Postscaler
	0010 = 1:3 Postscaler
	0011 = 1:4 Postscaler
	0100 = 1:5 Postscaler
	0101 = 1:6 Postscaler
	0110 = 1:7 Postscaler
	0111 = 1:8 Postscaler
	1000 = 1:9 Postscaler
	1001 = 1:10 Postscaler
	1010 = 1:11 Postscaler
	1011 = 1:12 Postscaler
	1100 = 1:13 Postscaler
	1101 = 1:14 Postscaler
	1110 = 1:15 Postscaler
	1111 = 1:16 Postscaler

bit 2	TMR2ON : Timer2 On bit
	1 = Timer2 is on
	0 = Timer2 is off
bit 1-0	T2CKPS<1:0> : Timer2 Clock Prescale Select bits
	00 = Prescaler is 1
	01 = Prescaler is 4
	1x = Prescaler is 16

TABLE 7-1: SUMMARY OF ASSOCIATED TIMER2 REGISTERS

Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0	Value on POR, BOR	Value on all other Resets
INTCON	GIE	PEIE	T0IE	INTE	RBIE	T0IF	INTF	RBIF	0000 000x	0000 000x
PIE1	—	ADIE	RCIE	TXIE	SSPIE	CCP1IE	TMR2IE	TMR1IE	-000 0000	-000 0000
PIR1	—	ADIF	RCIF	TXIF	SSPIF	CCP1IF	TMR2IF	TMR1IF	-000 0000	-000 0000
PR2	Timer2 Module Period Register								1111 1111	1111 1111
TMR2	Holding Register for the 8-bit TMR2 Register								0000 0000	0000 0000
T2CON	—	TOUTPS3	TOUTPS2	TOUTPS1	TOUTPS0	TMR2ON	T2CKPS1	T2CKPS0	-000 0000	-000 0000

Legend: x = unknown, u = unchanged, — = unimplemented read as '0'. Shaded cells are not used for Timer2 module.